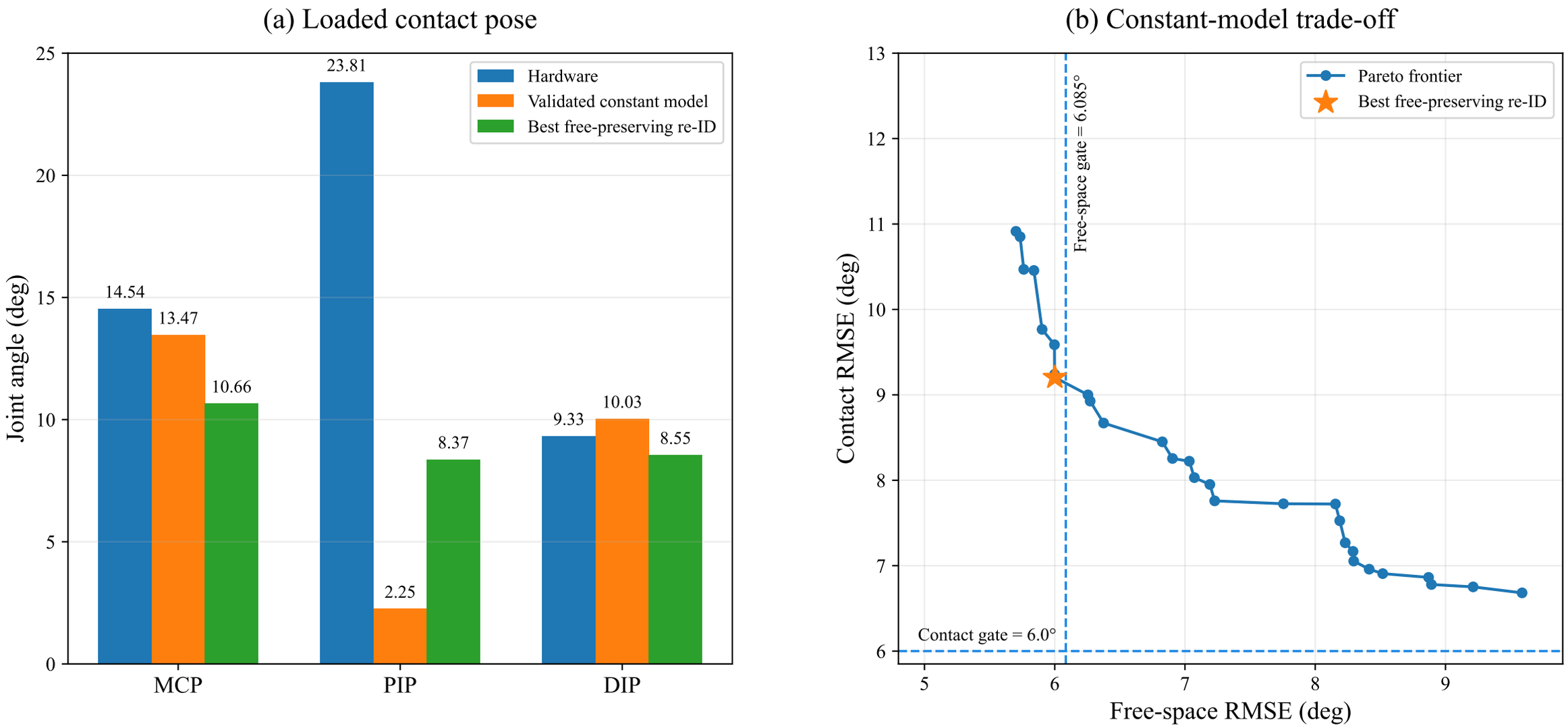
(a) Loaded contact pose
Joint angle (deg)
Hardware
Validated constant model
Best free-preserving re-ID
14.54
13.47
10.66
23.81
2.25
8.37
9.33
10.03
8.55
MCP
PIP
DIP
(b) Constant-model trade-off
Contact RMSE (deg)
Free-space RMSE (deg)
Pareto frontier
Best free-preserving re-ID
Free-space gate = 6.085°
Contact gate = 6.0°